

Gurjot Singh

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GitHub | LinkedIn | LeetCode | Codeforces

EDUCATION

• Gobindgarh Public School, Mohali <i>Class 11–12</i>	2021 – 2023 89%
• National Institute of Technology (NIT), Jalandhar <i>B.Tech. in Data Science and Engineering</i>	2023 – Present CGPA: 8.29

TECHNICAL SKILLS

- **Environment:** Windows, Linux, Docker, Docker-Compose, Kubernetes
- **Languages:** C++, C, JavaScript, TypeScript, Python, Golang
- **Web Development:** React.js, Next.js, Node.js, Express.js, tRPC, gRPC, WebSockets, Prisma ORM, SQL
- **AI & Machine Learning:** Supervised/Unsupervised Learning, Deep Learning (CNN, RNN, Transformers), Computer Vision, NLP, TensorFlow, Scikit-learn, OpenCV, LangChain, LangGraph, Agentic AI
- **Databases & Cloud:** PostgreSQL, MongoDB, MySQL, Firebase, Google Developer Console
- **Tools & Design:** Git, GitHub, Figma, Google Colab, Jupyter Notebook
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Database Management Systems, Natural Language Processing, Mathematics for Data Science, Computer Vision, Operating Systems

EXPERIENCE

- **CTP Portal — Backend Lead — NIT Jalandhar — 2024–Present:**
 - Built and maintained backend services for NITJ's placement portal serving **1000+ students** and **200+ recruiters**.
 - Developed **25+ secure REST APIs** for placements, verification, and job workflows.
 - Built role-based dashboards and optimized backend performance for peak placement seasons.**Tech Stack:** React.js, Express.js, MongoDB, TailwindCSS
- **Xceed Timetable Module — Top Contributor — NIT Jalandhar — 2025–Present:**
 - Revamped the timetable engine for conflict-free scheduling across **10+ departments**.
 - Implemented search, filtering, and Google Maps integration for **5000+ users**.
 - Optimized backend queries to improve scalability and response times.**Tech Stack:** React.js, Express.js, MongoDB, TailwindCSS
- **Third Space Travel — Full-Stack Developer — Freelance Project — 2026:**
 - Developed a travel booking platform with **30+ REST APIs** using React, TypeScript, Express, and MySQL.
 - Integrated the PhonePe Payment Gateway SDK for secure online payments.
 - Optimized database schema and queries using Prisma ORM for faster API performance.**Tech Stack:** React.js, TypeScript, Express.js, MySQL, Prisma ORM, tRPC, SCSS, Framer Motion, PhonePe Payment Gateway SDK

PROJECTS

- **Encrypted Chat System [Repo] — Sole Developer — 2026:**
 - Architected a highly reliable, end-to-end encrypted (E2EE) messaging platform capable of secure asynchronous chatting.
 - Implemented the **Double Ratchet Algorithm** for perfect forward secrecy in 1-to-1 chats and integrated **Sender Keys** to optimize group chat scaling.
 - Designed robust cryptographic handshakes utilizing public-key cryptography, alongside strict message integrity and multi-factor authentication methods.**Tech Stack:** Go (Golang), WebSockets, React.js, Protocol Buffers, PostgreSQL
- **Amazon Product Review Sentiment Analyzer — Course Project — 2026:**
 - Built an NLP pipeline using Python to preprocess, vectorize, and evaluate sentiment across extensive e-commerce review datasets.
 - Trained a **LinearSVC** classification model using TF-IDF vectorization, achieving high accuracy in distinguishing text polarity.
 - Generated comprehensive data visualizations using Matplotlib and Pandas to map out product feedback trend scores.**Tech Stack:** Python, Scikit-Learn, Pandas, Matplotlib, Jupyter/Colab

CERTIFICATES

- **Speed Quant (Competitive Programming) — Winner:** Ranked **1st** in the Main Track competing against top DSA programmers from 2nd and 3rd year at NIT Jalandhar. Recognized for speed and accuracy in solving algorithmic challenges under timed conditions.

CODING PROFILES

- **LeetCode:** Completed **700+** LeetCode problems showcasing strong DSA mastery and problem-solving skills.
- **Codeforces Specialist (1500+ Rating):** Solved **250+** problems and achieved a **1500+** rating across multiple contests, strengthening algorithmic thinking, data structures, and optimization skills through competitive programming.